

Seongjoon Moon

PhD Student
Korea University
Room 431, Innovation Hall, 145, Anam-ro, Seongbuk-gu, Seoul, Republic of Korea
janismoon@korea.ac.kr
+82) 10-6462-1927

EDUCATION

Mar. 2017 ~ Feb. 2023	Korea University Department of Mechanical Engineering	Seoul, Korea
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Advisor: Woojin Chung

Mechanical Engineering
GPA: 3.52 / 4.5

Mar. 2023 ~ Present	Korea University Department of Mechanical Engineering	Seoul, Korea
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Advisor: Woojin Chung

Ph.D. Student
GPA: 4.32 / 4.5

RESEARCH INTERESTS

- Reinforcement learning
- Imitation learning
- Mobile robot navigation
- Trajectory planning
- Multi-agent planning

JOURNAL PUBLICATIONS

1. Park, Seihoon, Jinwon Lee, Geonhyeok Park, Ikhyeon Cho, **Seongjoon Moon**, and Woojin Chung. "Robust Multi-Agent Path Finding Method for Obstacles and Environmental Changes in Factory Environments." *Sensors*, vol. 26, no. 13, 2026, article 4139. <https://doi.org/10.3390/s26134139>.

CONFERENCES

1. Seongjoon Moon, Boogeon Kim, Jinwon Lee, Geonhyeok Park, Ikhyeon Cho, Woojin Chung, "A Multi-Robot Navigation Framework in Large-Scale Indoor Workspace", The 19th Korea Robotics Society Annual Conference(KRoC 2024), Pyeongchang, Korea (Feb. 2024) - Oral

RESEARCH EXPERIENCES

- Research Student at School of Mechanical Engineering, Korea University, Korea (Feb. 2023 ~ Present) / Reinforcement Learning for Mobile Robot Navigation
- Research Student at School of Mechanical Engineering, Korea University, Korea (Jun. 2022 ~ Dec. 2024) / Multi-agent Path Finding and Task Allocation

SKILLS AND TECHNIQUES

- Languages: Python, C++
- Core: Reinforcement Learning (PyTorch), ROS/ROS2
- Expertise: Trajectory Generation, Multi-Agent Path Finding (MAPF), Multi-Robot Task Allocation (MRTA)
- Tools: Gazebo, Docker, Git, Linux